🔤 1. Text Detection Papers & Categories

A. Mathur et al. (2019) — Fast Regex Engine

***Model / Technique Used**: Pure JavaScript Regex in content.js*
***Goal**: Instant detection of high-frequency text patterns without waiting for AI.*

Category Name	Easy & Simple Explanation	How DarkSight Detects It
1. Urgency	Pressuring the user to buy fast using time countdowns or hurry text.	Matches phrases like `"offer expires"`, `"ends in"`, `"limited time"`.
2. Scarcity	Claiming inventory is almost gone to trigger fear of missing out (FOMO).	Matches phrases like `"only 2 left"`, `"low stock"`, `"selling fast"`.
3. Social Proof	Showing other shoppers' actions to make the item look popular.	Matches phrases like `"viewing right now"`, `"bought in last 24h"`.
4. Fake Discount	Misleading crossed-out price badges or exaggerated savings.	Matches phrases like `"99% OFF"`, `"Flat 50% discount today"`.

B. Schaefer et al. (2023) — Text LLM Taxonomy

***Model / Technique Used**: **DeepSeek-R1 / Mistral** via local Ollama API*
***Goal**: Deep psychological text classification across 8 academic categories.*

Category Name	Easy & Simple Explanation	How DarkSight Detects It
1. Misdirection	Distracting the user or using emotional guilt to influence choices.	Text LLM flags confirmshaming text like *"No thanks, I prefer paying full price"*.

2. Urgency	Manipulative time pressure beyond simple regex patterns.	Text LLM evaluates full paragraph context for artificial sales pressure.
3. Scarcity	Inventory pressure messages embedded in product descriptions.	Text LLM flags subtle artificial scarcity claims.
4. Social Proof	Testimonials or activity feeds designed to force quick decisions.	Text LLM detects unverified popularity claims.
5. Obstruction	Making cancellation, opt-out, or account deletion unnecessarily hard.	Text LLM flags phrases forcing phone calls or complex steps to cancel.
6. Forced Action	Forcing users to accept marketing consent or sign up to access features.	Text LLM flags mandatory agreement checkboxes bundled with access.
7. Sneaking	Hiding hidden fees, care costs, or automatic subscription add-ons.	Text LLM detects stealth charges embedded in terms or checkout summaries.
8. Fake Discount	Suspicious slash pricing or unrealistic discount bait banners.	Text LLM flags absurd discount claims (*e.g., "Was $5000, Now $50"*).

***Schaefer et al. (2024) Countermeasures Applied**:*
*• **Highlight (HLE)**: Wraps element in glowing red border and injects explanation tooltip.*
*• **Hide (HD)**: Sets element `display: none` to remove deceptive UI.*
*• **Switch (SW / Fair UI)**: Replaces pressure text (e.g., replaces "Hurry! 1 left" with "[Stock indicator hidden]").*

C. In-Context Learning for LLMs (2023) — Our DP In-Context Learning GPT

***Model / Technique Used**: **Few-Shot Prompt Engineering** in prompt.js*
***Goal**: Prevents LLM hallucinations by training it with positive & negative DOM text examples.*
***List of Dark Patterns Targeted**: Uses the same **8 Schaefer text categories** above, but provides real DOM text examples for each category inside the LLM prompt.*

📐 2. Structural Detection Paper & Categories

AidUI — Structural Visual Analysis Engine

***Model / Technique Used**: Pure JavaScript DOM Geometry (`getBoundingClientRect`) & CSS Color Contrast calculation in `content.js`*
***Goal**: Detect visual imbalance and unequal button prominent styling without AI.*

Category Name	Easy & Simple Explanation	How DarkSight Detects It
False Hierarchy (Attention Distraction)	Making the option the company wants (e.g., "Accept All Cookies") giant and brightly colored, while making the alternative (e.g., "Reject") tiny, greyed-out, or low-contrast.	DarkSight calculates button surface area (width × height) and background color contrast ratio. If discrepancy ≥ 1, it flags the element as a False Hierarchy.

👁️ 3. Visual Detection Papers & Categories

A. DP Sentinel — The 14 Dark Pattern Categories

***Paper Title**: Dark Pattern Sentinel: A Hybrid DOM Heuristics and LLM-Based Real-Time Detection System*
***Model / Technique Used**: `chrome.tabs.captureVisibleTab` + **Qwen2.5-VL / LLaVA** Multimodal Model via Ollama in `background.js`*
***Role in DarkSight**: Defines the 14 core dark pattern categories with severity levels, legal anchors, and scanner modules for the hybrid & visual pipeline.*

#	Dark Pattern Category	Severity Level	Easy & Simple Explanation	Scanner Module & Legal Anchor
1	**False Urgency**	HIGH (3)	Pressuring users to make fast choices using artificial deadline claims.	`urgency.js` (EU DSA Art. 25; CCPA 2023)

2	**False Scarcity**	**HIGH (3)**	Claiming inventory is almost gone to trigger fear of missing out (FOMO).	`scarcity.js` (FTC; CCPA Guidelines 2023)
3	**Manufactured Social Proof**	**MEDIUM (2)**	Showing unverified user activity or fake shopper notifications.	`social-proof.js` (FTC Fake Reviews Rule)
4	**Confirmshaming**	**HIGH (3)**	Using emotional guilt or shame phrasing on decline/opt-out buttons.	`confirmshaming.js` (EU DSA Art. 25)
5	**Hidden Costs / Drip Pricing**	**CRITICAL (4)**	Revealing extra fees, service charges, or delivery taxes late at checkout.	`hidden-costs.js` (CCPA §3; EU Omnibus)
6	**Sneak Into Basket**	**CRITICAL (4)**	Automatically pre-adding extra items, warranties, or protection into cart.	`sneak-basket.js` (EU CRD Art. 22)
7	**Pre-selected Consent**	**MEDIUM (2)**	Pre-checking checkboxes for marketing emails, data sharing, or upgrades.	`preselected.js` (GDPR Art. 7(2); DPDP 2023)
8	**Roach Motel**	**HIGH (3)**	Making it very easy to sign up or subscribe, but extremely hard to cancel.	`roach-motel.js` (FTC Click-to-Cancel 2024)
9	**Forced Continuity**	**HIGH (3)**	Silently charging recurring subscription fees after a free trial ends.	`forced-continuity.js` (FTC Negative Option)
10	**Disguised Ads**	**MEDIUM (2)**	Styling commercial advertisements to look like site menus or buttons.	`disguised-ads.js` (FTC .com Disclosures; ASCI)

11	**Trick Questions**	MEDIUM (2)	Using confusing double negatives or ambiguous phrasing on opt-out forms.	`trick-questions.js` (EU DSA Art. 25)
12	**Visual Interference**	MEDIUM (2)	Hiding decline links in tiny grey text or making accept buttons visually huge.	`visual-interference.js` (EU DSA; GDPR rulings)
13	**Fake Countdown Timer**	HIGH (3)	Displaying fake timers that reset on page refresh or never expire.	`countdown.js` (CCPA Guidelines 2023)
14	**Privacy Zuckering**	HIGH (3)	Tricking users into sharing more personal data than they intended.	`privacy-zuckering.js` (GDPR Art. 25; DPDP 2023)

B. 50 Shades of Deceptive Patterns — 21-Category Visual Vision Engine Taxonomy

***Model / Technique Used**: **Qwen2.5-VL / LLaVA** Multimodal Vision LLM prompt taxonomy in prompt.js*

***Goal**: Provides the 21 visual deceptive pattern definitions evaluated by the DP Sentinel Vision Pipeline.*

#	Visual Category Name	Easy & Simple Explanation	What DarkSight Detects
1	**False Hierarchy**	Primary button dominates the screen visually over decline options.	Visual layout comparison of button prominence.
2	**Disguised Ads**	Advertisements stylized to look like site menus or download buttons.	Image classification of ad banners mimicking native UI.

3	**Gamification**	Fake spin-the-wheel or scratch card popups to collect user data.	Detects interactive game popups over store pages.
4	**Price Comparison Prevention**	Layouts hiding item weights/unit prices to block cost comparison.	Detects missing/hidden comparative price badges.
5	**Hidden Costs**	Mandatory fees revealed visually only at the final payment step.	Identifies checkout screenshots with surprise fee additions.
6	**Sneak into Basket**	Extra products or protection plans pre-added into the cart visually.	Flags pre-populated cart items in screenshot.
7	**Roach Motel**	Visual design making sign-up prominent but cancellation hidden.	Identifies asymmetric user journey UI elements.
8	**Confirmshaming**	Decline buttons styled with emotionally manipulative visual cues.	Flags visual buttons containing guilt-inducing text.
9	**Trick Questions**	Confusing visual toggle switches with double negatives.	Detects ambiguous slider toggles.
10	**Nagging**	Repetitive intrusive popups blocking the main screen content.	Identifies modal overlays obstructing reading areas.
11	**Hard to Cancel**	Deep nested navigation menus for account cancellation.	Analyzes visual flow of settings screens.
12	**Pre-selected Options**	Checkboxes pre-filled for newsletter signups or insurance.	Detects checked visual box states on forms.

13	**Continuous Scroll**	Endless page loading designed to keep users scrolling infinitely.	Identifies infinite feed UI layouts.
14	**Visual Interference**	Hiding unsubscribe links in tiny grey text in page footers.	Identifies low-visibility link placement.
15	**Artificial Scarcity**	Fake visual badges claiming limited stock.	Identifies warning icon badges.
16	**Artificial Urgency**	Fake graphic countdown timer widgets.	Identifies animated timer graphics.
17	**Social Proof Popups**	Floating toast notifications showing live purchase activity.	Detects bottom-corner purchase notification boxes.
18	**Forced Registration**	Full-screen modal wall blocking content until user signs up.	Identifies backdrop blur overlays requiring registration.
19	**Hidden Subscriptions**	Recurring billing details hidden in small visual fine print.	Detects fine-print subscription disclosures.
20	**Defaulting**	Setting privacy-invasive choices as default visual selections.	Flags pre-selected privacy options.
21	**Deceptive Timers**	Timers that reset visually when the page refreshes.	Identifies static timer widgets.

🏰 4. Master Framework Paper Summary

Dark Pattern Sentinel (DP Sentinel)

***Model / Technique Used**: Multi-Tiered Hybrid Cascade Framework (content.js + background.js)*
***Goal**: Orchestrates all tools together. Tier 1 (Mathur Regex + AidUI DOM Geometry) runs instantly on the DOM. If content is complex, Tier 2 sends text to **DeepSeek-R1**, and Tier 3 (Multimodal Pipeline) captures a screenshot via `chrome.tabs.captureVisibleTab` and sends it to **Qwen2.5-VL**.*